Taichi Uchida

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I'm a fourth-year aerospace engineering student at Kyoto University. My research covers mission design, trajectory optimization, and robotics.

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Education

2027.04 **SOKENDAI · Planned enrollment**
Five-year doctoral program, Space and Astronautical Science. Graduate Institute for Advanced Studies. Naoya Ozaki's Mission Design Lab.

2023.04 - 2027.03 **Kyoto University**
Fourth-year undergraduate in Engineering Science (Aerospace Engineering). Graduation expected March 2027.

2020 - 2023 **Seiko Gakuin High School**
Kanagawa, Japan

2018 - 2020 **Pathways School Gurgaon**
IB Middle Years Programme · Haryana, India

2017 - 2018 **Seiko Gakuin Junior High School**
Kanagawa, Japan

Research & Development

2025.08 - **Outer-planet mission design and trajectory optimization**
JAXA / ISAS / Research intern · Naoya Ozaki Lab. Multi-point observations with small outer-planet probes, and low-thrust trajectory optimization with Julia and DDP.

2026.04 - **Model predictive control with a Transformer**
KYOTO UNIVERSITY / Undergraduate researcher · Kenji Fujimoto Lab. A Transformer-based dynamics model integrated with nonlinear MPC for real-time autonomous tracking on an RC car.

2025.10 - **Reinforcement learning and control for lunar robots**
KYOTO UNIVERSITY / Research assistant · Learning Machines Group. Reinforcement learning, inverse kinematics, and sim-to-real environments for modular multi-legged lunar robots.

2025.02 - 2025.04 **7-DOF robot arm control**
DFKI · BREMEN / Visiting research fellow · Robotics Innovation Center. Simulation environments and impedance / position controllers for a 7-DOF arm, with space-debris capture as the research context.

2025 - **Athletics record management web app**
KYOTO UNIVERSITY · TRACK & FIELD / Lead developer & system administrator. A React and Firebase app for the Kyoto University Medical School Track & Field Team, currently in trial use.

2025.03 - **Russian grammatical error correction**
TLEEZ INC. / Software engineer intern. AI-based grammatical error correction and training data generation for Russian language learning.

2024.08 - **Metagenomic analysis of microbial communities**
GEORGIA TECH / Visiting research fellow · DiChristina Lab. Metagenomic analysis of microbial communities involved in heavy metal contaminant stabilization.

2020.12 - 2023.03 **Isolation and identification of halophilic archaea**
TOKYO TECH · ELSI / Research intern · Kosuke Fujishima Lab. Isolation, identification, and phylogenetic analysis of halophilic archaea in rock salt from around the world.

Earlier Research

2023.09 - 2024.08	**Microbial metabolism · Kyoto University** Wet-lab research on hydrogen and oxygen utilization at the Atomi Lab.
2021.08 - 2022.02	**Meteorite analysis · Seiko Gakuin** SEM-based composition analysis with UK schools and university researchers.
2020.08 - 2021.08	**Artificial gravity · Seiko Gakuin** Prototyped a mechanism and built Arduino-based force sensors for evaluation.
2021	**Intelligent water rocket** A rocket with computer-controlled functions and automated parachute deployment.

Presentations & Papers

2026.08	**HDDP-based trajectory optimization for low-thrust GTO-to-GEO transfers** 70th UKAREN. Paper submitted.
2026.03	**Mission design for multi-point simultaneous observations by ultra-small outer-planet probes on large overseas missions** SPS 2026. Oral presentation.
2023.11	**Space medicine education in Japan** 69th JSAEM. Oral & poster.
2022.06	**Extreme halophiles from global rock salts** NASA × AGI × High School Students. Online presentation.
2022.05	**Isolation and identification of extreme halophiles from global rock salts** JpGU 2022. Poster · Award.
2022.03	**Research on extreme halophiles** Japanese Society of Scientific Fisheries. Online presentation.
2022.01	**Meteorite composition analysis with electron microscopy** International Research for School. Online presentation.

Awards & Scholarships

2025	**Startup Weekend Space Osaka · 2nd place** Team leader; business plan and prototype in 54 hours.
2024-25	**Nakatani Foundation Scholarship** Fully funded research at Georgia Tech and DFKI.
2022	**JpGU Award for Academic Excellence** Research on halophilic archaea.
2022	**Japanese Society of Scientific Fisheries Award** Extreme halophiles research.
2022	**SDGs Tankyu Award · Nihon Ryoko Prize** Founded and organized Seiko SDGs Food Ambassador.
2021	**Intelligent Water Rocket · Missions Award** Completed all competition missions.
2021	**SSH Research Symposium Award** Artificial gravity research.
2021	**Yurikamome Relay Marathon · Overall winner** Team competition.
2019	**ISSO National Games · Three golds** 1st in 800m, 1500m, and 3000m in India.

Activities

2026.09.15	**OpenAI Codex Student Builder Fest 2026** Developed WalkIsFun, a map service for finding the shortest route through shade, in a three-person team at OpenAI's student hackathon. Served as project manager.
2026.03	**JENESYS2025 · Japan-Korea exchange** Selected delegate for an international cultural exchange program.
2025.05	**Interpreting at the Kyoto International Music Students Festival** English-Japanese interpreter and liaison at the Kyoto International Music Students Festival.
2023 -	**Mathematics instructor at Tetsuryokukai** Advanced mathematics instructor at Tetsuryokukai.
2021	**Japan Students Space Forum** Founded and organized a forum with more than 100 participants.
2021	**Seiko SDGs Food Ambassador** Founded a project supporting regional producers through a school festival.
School years	**The "Yokohama Banana" project** Founded and managed a venture that sold 3,800 chocolate bananas over two days.
2025 / 2023	**Space medicine & Moon Village** Staff at the 71st JSAEM and 7th Global Moon Village Workshop; organized a university space medicine study tour.
2012 -	**Rice farming & Shorinji Kempo** Long-term participation in organic rice farming; competed at the national junior-high Shorinji Kempo championship.
2021 / 2019	**Science & international exchange** Founded the Science Club Summit; organized India-Japan student discussions and a Japanese calligraphy event.
2021	**Legislative internship** ICAS program at the Tokyo Metropolitan Assembly.

Skills & Qualifications

Programming Python · Julia · C++ · MATLAB · R · Go

Control & robotics DDP · NMPC · Drake · Pinocchio · MuJoCo · Isaac Lab · CasADi · ROS2

Machine learning & web development PyTorch · React · Next.js · Firebase · Docker · Linux · Git

Life science PCR · DNA extraction · PhyML · Metagenomics

Languages Japanese (native) · English (TOEFL iBT 100 / EIKEN Grade 1) · Russian (learning)

Qualifications Model Rocket License Grade 4 · NAUI Open Water Diver · Driver's license (AT)